

Rohan Jain

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SUMMARY

I build LLM systems that hold up in production, not just in demos. At Bright Machines I own the agent layer of an on-prem AI platform: LangGraph agents calling 132 MCP tools through a LiteLLM gateway, plus the FastAPI and React/TypeScript services around them on Kubernetes. I measure before I believe: I killed my own recommended design when it scored 0.03 recall, and traced a 74-minute outage that every health check called healthy. MS Data Science, CU Boulder, GPA 3.9, May 2026.

SKILLS

LLM & Agents: Python, LangGraph (ReAct, multi-agent), LangChain, Model Context Protocol (built and hardened MCP servers), LLM tool-calling agents, RAG (BM25, embeddings, hybrid), Pydantic v2 structured outputs, LiteLLM multi-provider gateway, OpenAI / Anthropic / Gemini APIs, LLM evaluation harnesses, prompt-injection testing, guardrails

Backend & Data: FastAPI, PostgreSQL + Alembic, SQL, SQLite / sqlite-vec, Databricks (Unity Catalog, Unity AI Gateway, Model Serving, Genie, AI Functions; Iceberg CDC via Debezium), Temporal, Docker, Kubernetes / Kustomize, ArgoCD, GitLab CI

Frontend, Cloud & Observability: TypeScript, React, Next.js, Tailwind; Azure (AKS, Entra ID), AWS (EC2, Lambda, S3, API Gateway, RDS); OIDC auth (oauth2-proxy, Keycloak); Langfuse tracing, Grafana, Prometheus, MLflow

WORK EXPERIENCE

AI Engineering Intern - Bright Machines

Jun 2026 – Present | San Francisco, CA

- Owned tool selection for a production on-prem LLM agent (LangGraph ReAct over a LiteLLM gateway; 132 MCP tools spanning Postgres, Kubernetes, Grafana and Temporal against a 32K-token window); built a 115-question evaluation harness that raised tool recall from 0.47 to 0.97 in ~80 lines.
- Designed and shipped a read-only-by-construction MCP server from scratch (37 files, +3,163 lines, 12 tools) serving a 32,000-workflow Temporal namespace in production; Docker-packaged, deployed via Kubernetes/Kustomize, every mutation path removed and CI-gated.
- Hardened the agent's four MCP servers: closed a Critical path-traversal vulnerability (CWE-22), enforced least-privilege RBAC, replaced a fail-open deny-list with a fail-closed allow-list, and proved via adversarial prompt injection that the agent cannot execute database writes.
- Root-caused a 74-minute silent chat outage (agent registry not rehydrating after a gateway restart while health probes stayed green), disproved the planned upgrade fix by inspecting the running container, and shipped Postgres persistence plus a self-healing watchdog (135 tests), verified live in production.
- Built an on-prem grounded document Q&A platform end to end (FastAPI backend, React/TypeScript UI, OpenAI-compatible endpoint, sqlite-vec retrieval, fail-closed OIDC identity seam behind oauth2-proxy) and deployed it to Azure AKS via ArgoCD with 350+ tests; held the release until sign-in existed rather than expose it publicly.
- Scoped AI adoption on the production Databricks lakehouse (Unity Catalog; Iceberg CDC from Postgres via Debezium): found Databricks AI had never been enabled, mapped Unity AI Gateway and Model Serving onto the team's LiteLLM/Ollama stack, and delivered a ranked proposal (Genie over the silver layer, AI Functions, MLflow evals).

Data Science & Engineering Intern - Nexus Weather & Climate

May 2025 – May 2026 | Boulder, CO

- Replaced a legacy SVM bias-correction model with a tuned Random Forest for NWP post-processing (feature engineering, cross-validation), cutting temperature MAE 47% and humidity MAE 54%; promoted to production.
- Built an automated inference quality gate rejecting any model that fails to beat the pre-ML baseline, cutting failed production runs 90% and saving ~20 engineer-hours per month.

Software Developer - Kopf Lab, University of Colorado Boulder

Dec 2025 – May 2026 | Boulder, CO

- Migrated lab infrastructure to AWS EC2 behind Keycloak OIDC with group-based access control and Grafana/Loki/Prometheus observability; reverse-engineered proprietary Thermo Isodat binaries into deterministic R readers with golden-output regression tests.

EDUCATION

University of Colorado, Boulder

Master's in Data Science (GPA: 3.9/4.0)

Aug 2024 – May 2026 | Boulder, CO

Mumbai University, Rajiv Gandhi Institute of Technology

B.E. in Computer Engineering (GPA: 3.7/4.0)

Jul 2020 – May 2024 | Mumbai, India

PROJECTS

ClaimPilot AI (Guardrailed Tool-Calling Agent) | Python, Pydantic v2, OpenAI function calling

Jan 2026

- Deterministic Pydantic v2 checks surface claim issues while OpenAI function calling proposes structured fixes; strict output schemas with retry/fallback guarantee a valid report even when the model fails.

AgentSquared (No-Code AI Agent Builder) | FastAPI, SQLAlchemy, Next.js, Tailwind, Gemini

Mar 2026

- HackCU 12: FastAPI + SQLite backend and Next.js/Tailwind frontend where every agent is a database row plus JSON config; Gemini RAG over uploaded knowledge bases plus a Bluesky integration with human-approved replies.